Eisengrau

Making GB electricity data easier to investigate.

GB energy | algorithms | analysis

Find patterns. Test ideas.

Matthew, Mark, David

ONE PRICE.
SEVERAL MOVING PARTS.

When imbalance prices move,
three drivers shift.

We track them independently
at the horizon that matters.

- Imbalance price
- Wind forecast and available outturn
- Demand forecast and available outturn



OVERVIEW.

THE SHAPE OF A MARKET DAY.

One settlement day at a time

Forecasts stay distinct from actuals

Missing values remain visible

Settlement days have 46, 48 or 50 half-hour periods

PRICE	WIND	DEMAND
£/MWH	GW	GW
Half-hourly	Half-hourly	Half-hourly

Source: Elexon | Data freshness: Intraday



TERMINAL

Turn a question into an investigation.

Linked charts and tables keep every view in sync.

Saved workspaces and reusable presets save time.

CSV exports make data easy to share.



THE BEST MODEL
DEPENDS ON THE HORIZON.

In one 53-day Tiger dev sample from 1 Sep to 23 Oct 2026 (2,544 half-hour periods; 28-day rolling refit)

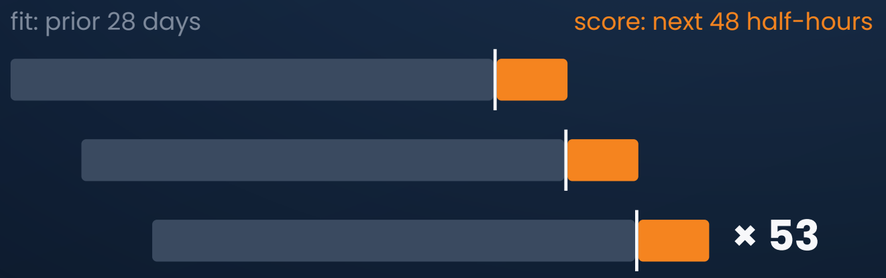
30 MINUTES AHEAD	24 HOURS AHEAD	48-PERIOD AVERAGE
£19.2 /MWh	**£39.4 /MWh**	**£49.5 /MWh**
naive / drift MAE	28-day mean MAE	ETS average MAE

The winner changes with lead time.

These results reflect one autumn sample, not a guarantee.



LOWER ERROR.
OR JUST NOISE?



white tick = origin, 00:00 London; refit from scratch at every origin

WALK-FORWARD

IN APP

$$\hat{\theta}_k = \mathrm{fit}\left(y_{[o_k-1344,\ o_k)}\right) \Rightarrow \hat{y}_{[o_k,\ o_k+48)}, \qquad o_{k+1} = o_k + 48$$

Refit each midnight on the prior 28 days; targets never enter the fit.

53 midnight origins: each lead is one clock slot

SCALED ERROR (MASE)

IN APP

$$\mathrm{MASE} = \frac{1}{53}\sum_{k=1}^{53} \frac{\mathrm{MAE}_k}{\overline{|y_t - y_{t-48}|}_{\text{ prior 28 days}}}$$

Each day's error over the prior 28 days' typical day-on-day change: unit-free.

ETS 0.84 · next best 0.89 · same slot yesterday 1.10

DIEBOLD–MARIANO

IN ENGINE

$$\mathrm{DM} = \frac{\bar{d}}{s_d/\sqrt{n}} \overset{H_0}{\approx} t_{n-1}, \qquad d_k = \mathrm{MASE}_k^{\mathrm{ETS}} - \mathrm{MASE}_k^{\mathrm{next}}$$

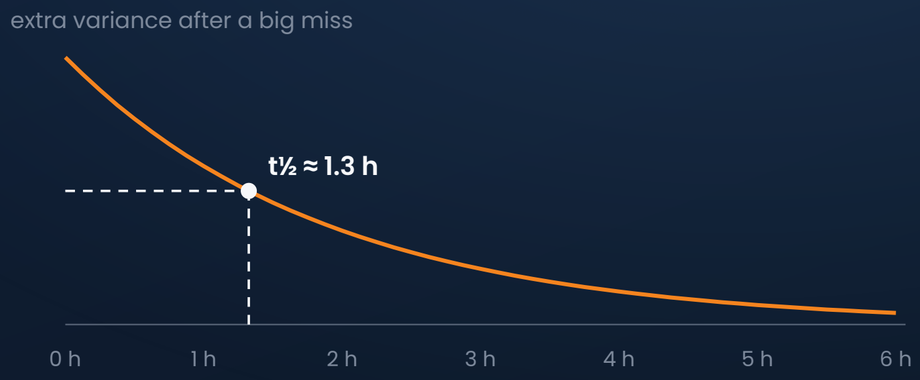
Is the daily gap bigger than day-to-day noise? Days don't overlap, so h = 1.

p = 0.02 over 53 days: unlikely to be noise

IN APP runs live in Eisengrau IN ENGINE coded and tested, not yet live

Elexon imbalance price · 1 Sep–23 Oct 2026 · 2,544 half-hours · slide 5's sample

ERRORS CLUSTER. CONSTANT-VARIANCE BANDS MISLEAD.



INTERVAL SCORE

IN APP

$$\mathrm{IS}_{80} = \frac{1}{n}\sum_t \left[(u_t - l_t) + 10\,(l_t - y_t)^+ + 10\,(y_t - u_t)^+\right]$$

Charges band width plus misses; coverage alone rewards bands that are too wide.

IS_{80} £/MWh: naive 508 (99% inside) vs mean 260 (76%)

ARCH-LM TEST

IN APP

$$e_t^2 = a_0 + \sum_{j=1}^{12} a_j\, e_{t-j}^2 + \eta_t, \qquad \mathrm{LM} = (n-12)\,R^2 \overset{H_0}{\approx} \chi^2_{12}$$

Do big misses follow big misses within 6 h?

LM = 196 vs 21 at 5%; 219 with time of day removed

GARCH(1,1)

IN ENGINE

$$\sigma_t^2 = \omega + \alpha_1 e_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \qquad t_{1/2} = \frac{\ln 2}{-\ln(\alpha_1+\beta_1)}$$

Variance jumps after a big miss, then decays; the half-life says how fast.

Half-life ≈ 1.3 h, time of day removed (derived)

IN APP runs live in Eisengrau IN ENGINE coded and tested, not yet live

Elexon imbalance price · 1 Sep–23 Oct 2026 · 2,544 half-hours · slide 5's sample

WHAT PRICE REMEMBERS.

Next: residual-demand forecasts as model inputs, versioned at each origin.



AUTOCORRELATION

IN APP

$$\mathbb{E}\,(y_t - y_{t-k})^2 = 2\sigma^2(1-\rho_k) \;<\; \sigma^2 \iff \rho_k > \frac{1}{2}$$

Derived: the last price beats the 28-day mean only while ρ > ½.

0.81 at 30 min, 0.31 at 24 h: slide 5's winners

KALMAN FILTER

IN ENGINE

$$y_t = \mu_t + \gamma_t + \varepsilon_t, \quad \mu_{t+1} = \mu_t + \eta_t, \quad z_t = \frac{y_t - \hat{y}_{t|t-1}}{\sqrt{F_t}}$$

Splits price into a drifting level, a daily cycle and noise; z measures surprise.

|z| > 3 in 1.3% of half-hours; a normal gives 0.27%

DISTURBANCE SMOOTHER

IN ENGINE

$$r_t^{\varepsilon} = \frac{\hat{\varepsilon}_t}{\sqrt{H - \mathrm{Var}(\varepsilon_t \mid Y_n)}}, \qquad r_t^{\eta} = \frac{\hat{\eta}_t}{\sqrt{Q - \mathrm{Var}(\eta_t \mid Y_n)}}$$

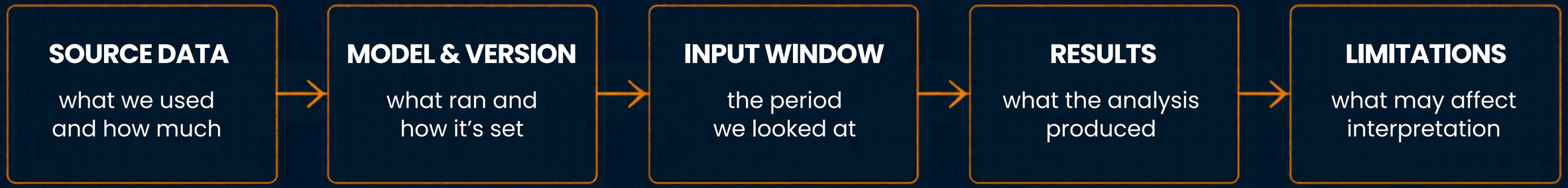
In hindsight: one-off spike (ε) or lasting level shift (η)? Upstream denominator fixed.

20 spikes, 40 shifts flagged; ~7 each if the model held

IN APP runs live in Eisengrau IN ENGINE coded and tested, not yet live

Elexon imbalance price · 1 Sep–23 Oct 2026 · 2,544 half-hours · slide 5's sample

EVIDENCE TRAVELS WITH EVERY RESULT.



Missing is not zero.

Absence of data is not evidence of absence.

A correlation is not a cause.

Observed relationships are not causal claims.

Retropective is not knowable.

Past analysis reflects what we saw, not what could have been known.

FROM SOURCE DATA TO A SAVED INVESTIGATION.

A clear pipeline turning raw market and weather data into actionable analysis you can trust.

SOURCE RECORDS
Elexon and NESO market and system data

→

INGESTION
Backend ingestion and quality checks

→

ANALYSIS ENGINE
Tiger Data and documented algorithms

→

API
Structured access to analysis results

→

APPLICATIONS
Eisengrau Overview and Terminal (hosted on Vultr)

NEXT EXTENSIONS

Evidence-linked Gemini analysis

Synoptic weather map for deeper context



FOLLOW
ONE MARKET DAY.



01 → **EXPLORE THE DAY**

See the market move from open to close.

02 → **COMPARE THE METHOD**

Test performance at a specific lead time.

03 → **FOLLOW THE EVIDENCE**

Inspect the data behind the result.

IMBALANCE INTELLIGENCE | Matthew · Mark · David